DO NOW — RECALL

BIOS, drivers and virtual machines

Name: _____ Class: _____ Date: _____

1. What does an OS driver do?

2. Define multi-user OS.

3. What executes before the OS is loaded?

4. Simplify (¬A∧B)∨(A∧B). Show factoring.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Interfaces with specific hardware.

2. Model answer

Supports separate user sessions and permissions.

3. Model answer

Platform firmware and boot-loading code.

4. Model answer

B∧(¬A∨A)=B∧1=B.

Vocabulary — know these terms

UEFI — Modern system boot firmware.

driver — Hardware-specific software interface.

VM — Virtual machine.

hypervisor — VM hosting software.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Model and check

Explain why each value follows from the earlier step.

IMPROVE: Explain one correction to your own reasoning.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A teacher needs to test a new OS without replacing their existing install. Choose a solution.

A VM provides a reusable guest environment with snapshots, although it consumes host resources.



WEDO · STEP 2

NEW LEARNING

Read, annotate and explain

Understand

BIOS/UEFI firmware initialises key hardware and starts the boot process.

Trace

Device drivers connect OS abstractions to the instructions and data expected by actual peripherals.

Evaluate

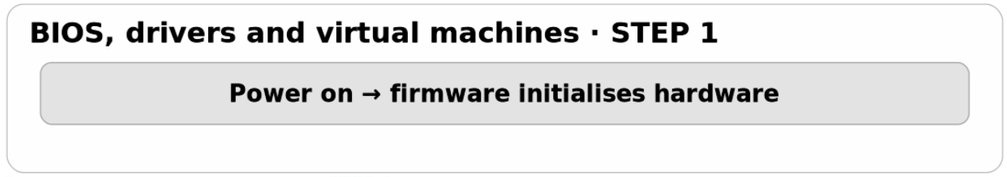
A hypervisor provides virtual machines for isolation, testing and consolidation, though CPU/RAM/storage overhead remains.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

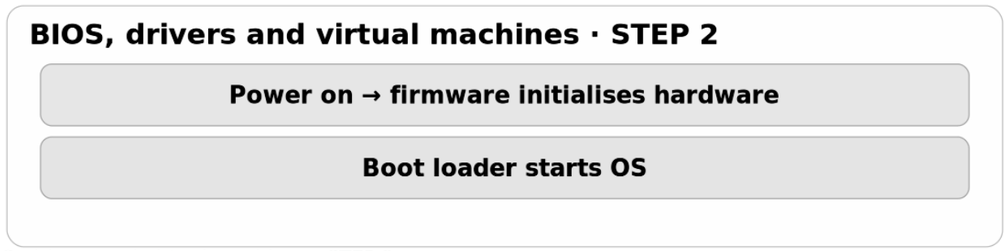
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



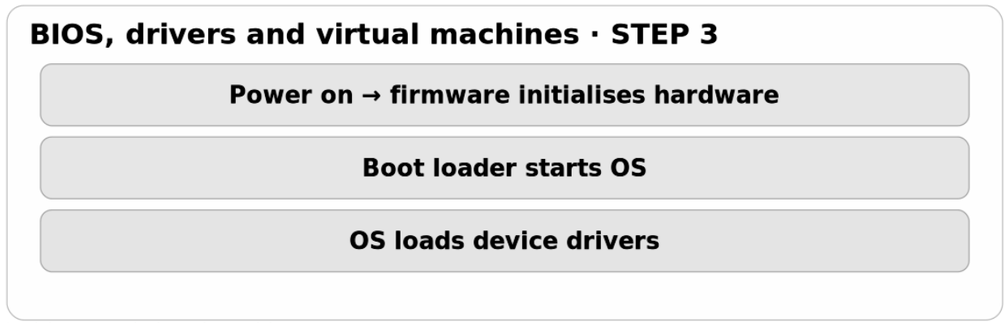
BIOS, drivers and virtual machines · STEP 1

New Learning



BIOS, drivers and virtual machines · STEP 2

New Learning



BIOS, drivers and virtual machines · STEP 3

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A virtual machine has its own wholly separate physical processor.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

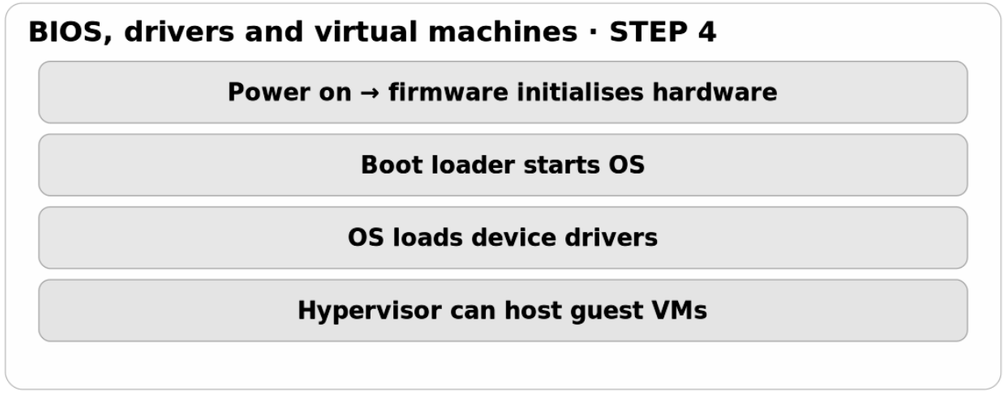
Question 3

Evaluate guest virtual machines for school testing of unfamiliar operating systems. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



BIOS, drivers and virtual machines · STEP 4

I DO — WORKED EXAMPLE

Follow the worked method

A printer is connected but software cannot print; name an OS dependency.

Predict first

A missing or incompatible driver can prevent correct communication with a device.

Model and check

Explain why each value follows from the earlier step.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

State roles of UEFI/BIOS, boot loader, driver and hypervisor. [4]

Question 2

Explain how a compatible printer driver enables printing. [6]

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first A VM provides a reusable guest environment with snapshots, although it consumes host resources.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

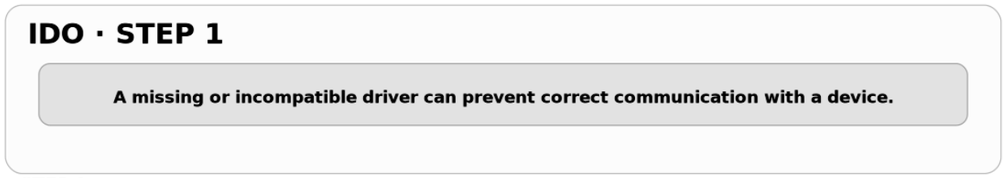
IMPROVE — Write one precise correction or stronger explanation in your own answer.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A printer is connected but software cannot print; name an OS dependency.

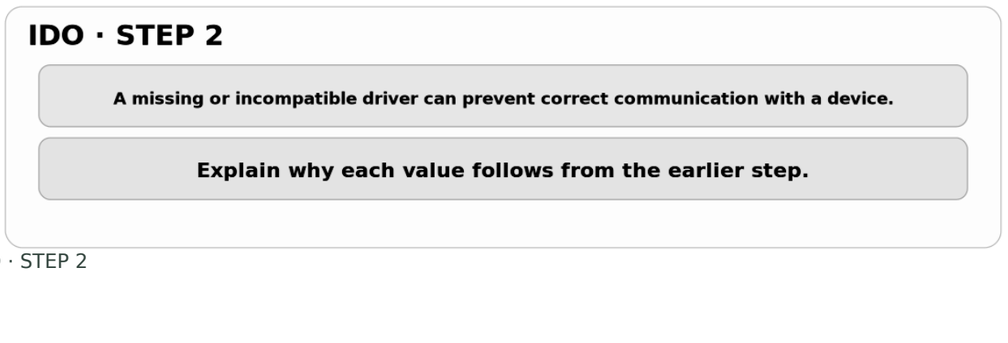
A missing or incompatible driver can prevent correct communication with a device.



IDO · STEP 1

A printer is connected but software cannot print; name an OS dependency.

A missing or incompatible driver can prevent correct communication with a device.



IDO · STEP 2

WE DO — GUIDED PRACTICE

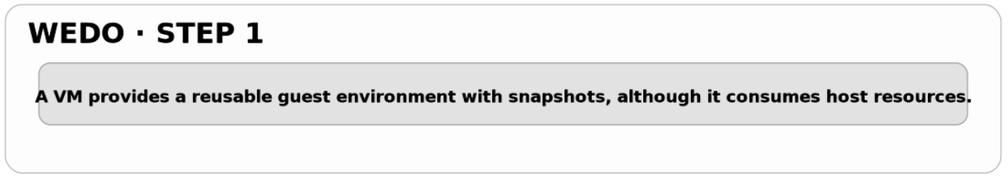
Try → Discuss → Improve

We Do · original lesson diagrams

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WEDO · STEP 1